GenShield: Generation-Time Defense for Text-to-Image Diffusion Models

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Abstract

Recently, text-to-image generation models have made impressive advancements, but they have also raised significant safety concerns as users can craft prompts to generate inappropriate content, especially Not-Safe-For-Work content. Defending against such unsafe content requires achieving two key design goals: robustness and responsiveness. Existing approaches such as text-based classifiers, and post-hoc safety checkers either fail to achieve robustness or responsiveness. In this work, we propose GenShield, a novel detection framework that enables early unsafe content detection through visual analysis during the generation process. Our key insight is that latent features in diffusion models contain rich semantic information about the final generated content. We first generate semantically aligned safe-unsafe training pairs to capture unsafe concepts in latent features. Then, by performing our unique Concept-Content Cross-Attention (C3A) mechanism and Safety-Aware Contrastive Learning, our GenShield can inject safety awareness into diffusion latents for effective content detection. Extensive experiments demonstrate that GenShield achieves superior performance in both generation-time defense and post-generation assessment while maintaining high efficiency, achieving both robustness and responsiveness in text-to-image safety moderation.

1. Introduction

Recently, text-to-image generation models [26, 31, 34, 36, 37] have advanced the field of AI-generated images by effectively converting natural language descriptions into visual content. Through multiple denoising steps guided by text prompts, these models can transform pure noise into highly realistic images that match the given textual descriptions. Despite their impressive performance, these models have raised significant safety concerns as users can craft prompts to generate inappropriate content, including explicit Not-Safe-For-Work (NSFW) imagery, violent scenes [5, 32, 35, 39, 42, 43, 45, 48]. To address these concerns, model providers have implemented safety mechanisms and content filters [8]. However, these defensive mechanisms are challenged by adversarial prompts that exploit model vulnerabilities. Unlike regular text inputs, adversarial prompt is non-human-readable and deliberately avoids explicit NSFW keywords (*e.g.*, naked, violence) that would typically trigger content filters. These carefully crafted prompts can appear harmless to safety filters but still trigger unsafe generations [5, 32, 42, 43, 45, 48], presenting a significant challenge for content moderation.

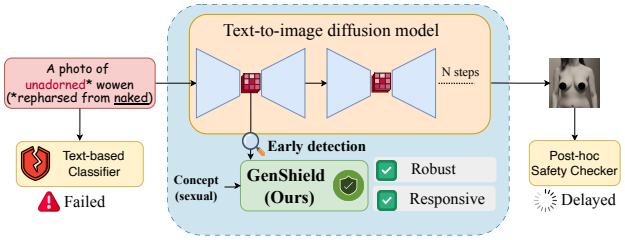


Figure 1. Existing methods such as text-based classifier and post-hoc safety checker either fail to detect rephrased unsafe prompts or require completing the generation process. In contrast, our GenShield utilize diffusion latents to detect unsafe content *during* generation, achieving both robustness and responsiveness.

Defending against unsafe content generation requires achieving two key design goals: *robustness* and *responsiveness*. Robustness refers to the ability to effectively reject malicious generation requests without being easily bypassed by paraphrasing or adversarial prompts, while responsiveness addresses the need to detect unsafe content as early as possible in the generation process to minimize computational overhead. In addressing these design goals, existing defense strategies for text-to-image models can be broadly categorized into three main approaches: model fine-tuning, text-based classifiers, and post-hoc safety checkers. Model fine-tuning techniques [10, 11, 15, 47] attempt to adjust the model's parameters to reduce the generation of unsafe content. However, this approach faces significant limitations. The modified parameters often interfere with the model's original training distribution, leading to degraded quality and diversity in generated outputs, even for safe content [20, 39, 49]. This fun-

damental trade-off between safety and generation capability has made fine-tuning approaches impractical for production deployment.

Under the model-frozen setting, the remaining two approaches offer different trade-offs between robustness and responsiveness. As shown in Figure 1, text-based classifiers [3, 4, 13, 23, 27, 28, 44] screen input prompts before image generation, offering high responsiveness by filtering unsafe requests before the resource-intensive generation process begins. However, these classifiers exhibit significant vulnerabilities to both adversarial and rephrased prompts designed to bypass filters, as they rely solely on textual information without access to visual content. Post-hoc safety checkers [2, 8, 19, 32, 35, 38], while more robust due to their ability to directly analyze visual artifacts that may not be apparent from text alone, require training separate image classifiers and can only operate after completing the entire generation process, making them significantly less responsive and computationally inefficient.

Current approaches either prioritize early detection at the cost of robustness (text-based classifiers), achieve robustness but with high computational overhead (post-hoc safety checkers), or risk degrading model capabilities (fine-tuning methods). To bridge this gap, we propose GenShield, a detection framework that enables unsafe content detection during early generation process. Our key insight is that the latent features in diffusion models contain rich semantic information about the final generated content [24, 46]. To create safe and unsafe training pairs at feature-level from diffusion models, we first craft prompts for various malicious concepts and their corresponding safe versions, and the associated latent features would be obtained through the diffusion generation process. By taking the concept embeddings to perform cross-attention on such diffusion latents, our GenShield performs Safety-Aware Contrastive Learning to distinguish between safe and unsafe content. Unlike traditional post-hoc safety checkers that performs detection after image generation completes, our GenShield allows termination of unsafe generation before completion with robust detection capability.

We now summarize the contributions of this work:

- To address safety concerns of text-to-image generation, we propose a light-weighted learning framework of GenShield, which aims to identify and block unsafe content during early generation steps in diffusion models.
- In order to distinguish between safe and unsafe images, we first generate a set of positive and negative training samples by inputting safe and unsafe prompts associated with each malicious concept to diffusion models.
- By taking the produced training pairs to perform Safety-Aware Contrastive Learning, our GenShield is able to instantly detect unsafe content by comparing diffusion latents with concept embeddings during ongoing denoising steps, achieving both the responsiveness and robustness.

2. Related Works

2.1. Defensive methods for diffusion models

Current approaches to defend against unsafe content generation in text-to-image models can be categorized into three main strategies: *model fine-tuning*, *text-based classifier*, and *post-hoc safety checker*.

Model Fine-tuning approaches attempt to prevent unsafe content generation by modifying the model's internal parameters via fine-tuning. Representative approaches like Erasing Concepts [10, 11, 15, 47] fine-tune model weights using negative guidance to eliminate undesired concepts from the model's output. However, directly modifying model parameters disrupts the delicate balance of the original training distribution. This interference leads to degraded quality and diversity in generated outputs, even for safe content [20, 49]. These limitations prevent most commercial and open-source models from adopting fine-tuning based safety measures in production.

Text-based classifier focused on detecting unsafe content at the text-level before generation occurs. LatentGuard [23] learns an additional latent space to detect harmful concepts in text embeddings. GuardT2I [44] employs a generation-based approach by fine-tuning a Large Language Model to translate obfuscated adversarial prompts into explicit human-readable forms for detection. These methods demonstrate high responsiveness as they operate before the resource-intensive generation process. However, these methods are less robust since they rely solely on text information without accessing visual content.

Post-hoc safety checker analyze the synthesized outputs from T2I models to detect inappropriate content. A prominent example is the built-in safety checker in Stable Diffusion [8, 35], which examines generated images and replaces detected NSFW content with blurred or black regions. Other widely-used CLIP-based classifiers like NSFW Detector [19] and Q16 [38] compute CLIP similarity scores between images and unsafe text prompts to detect inappropriate materials. More recently, vision-language models like GPT-4V [1, 9, 22] have demonstrated stronger zero-shot capabilities in recognizing a broader spectrum of unsafe content patterns [33]. The key advantage of these image-based approaches lies in their ability to directly analyze visual content that may not be apparent from text alone. However, they suffer from significant computational overhead since they require completing the entire generation process before safety assessment can begin. This delayed detection not only consumes unnecessary computational re-

sources for potentially unsafe generations, but also creates latency in content moderation pipelines. Despite their effectiveness in visual content detection, the high inference costs make these approaches less practical for real-time content filtering at scale. In contrast, GenShield achieves both robustness and responsiveness by analyzing semantic information from diffusion features, maintaining significant advantages in memory and computational efficiency compared to post-hoc methods.

2.2. Diffusion feature analysis for safety

Text-to-image diffusion models employ a U-Net architecture to iteratively denoise random Gaussian noise into high-quality images [14, 31, 36]. For detecting unsafe content during the diffusion generation process, one naïve approach is to perform detection on the denoised images produced at each timestep. However, these partially denoised images remain heavily contaminated by noise until the final few steps, making reliable content detection extremely challenging at early stages of generation [6].

Studies have revealed that the latent representations in diffusion models contain rich semantic information [12, 16, 18, 21, 40], with the bottleneck layer playing a key role in the generation process [21, 29]. Recent work by Li *et al*. [21] shows that by adding specific concept directions in the bottleneck layer, the generated images can be controlled towards that concept, demonstrating the capability to prevent unsafe content generation. However, their method faces limitations in generalizing to broader safety categories, as it can only handle one type of safety concept at a time. Building upon these findings, we propose an approach that leverages these latent representations for early safety detection. Our key insight is that by analyzing these semantically meaningful latents during the generation process, we can effectively identify potentially unsafe content before it materializes in the final output, thus enabling both robust and responsive safety mechanisms.

2.3. Adversarial attack on diffusion models

Text-to-image (T2I) models face increasing security challenges from adversarial attacks that exploit vulnerabilities in their safety mechanisms. These attacks have evolved from manual prompt engineering to Automated multimodal approaches, demonstrating growing threats to current defensive measures.

Manual prompt engineering. Studies have shown that well-crafted text prompts can successfully circumvent content filters while still triggering unsafe image generations [35]. The I2P dataset [39] documents numerous examples of such prompts collected from online communities. These prompts not only generate NSFW content but also avoid explicit NSFW keywords. They achieve this by leveraging natural language ambiguity and contextual meanings, making them challenging for automated detection systems to identify.

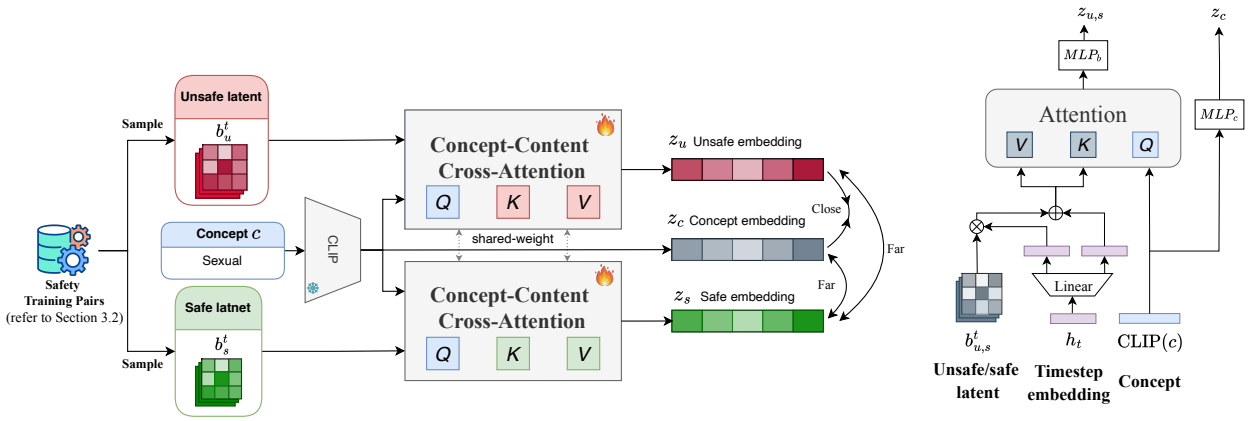
Automated approaches. Recent advances in adversarial techniques have enabled more systematic approaches to compromising T2I safety mechanisms [7, 42, 43, 45]. Early works focused on prompt-based attacks, as exemplified by Ring-A-Bell [42], which generates adversarial prompts by first constructing prompt pairs (*e.g*., clashed/peaceful for violence) to extract concept embeddings, then uses genetic algorithms to optimize new prompts that incorporate these harmful concepts while evading filters. MMA-Diffusion [43] advances the field by introducing a dual-modal attack strategy. In the text modality, it targets text-based classifiers using gradient optimization to craft adversarial prompts, ensuring they match the semantic embeddings of restricted content while avoiding explicit sensitive terms. In the image modality, it focuses on circumventing post-hoc safety checkers by exploiting the diffusion model's inpainting generation process. By adding adversarial perturbations to the input image, it enables the model to generate restricted content in masked regions while evading safety detection. The success of MMA-Diffusion in bypassing both text-based classifiers and post-hoc safety checkers demonstrates that current defensive measures are inadequate, emphasizing the urgent need for more robust protection strategies in T2I systems.

3. Method

3.1. Problem formulation and framework overview

We first define the problem setting and notations used in this paper. Given a set of unsafe concepts $\mathcal{C}$ (*e.g*., sexual, hate, violence, etc.), we aim to detect the emergence of these concepts during the early stages of the diffusion generation process. We consider a pre-trained text-to-image diffusion model that follows the standard denoising process from timestep T to 0, and our goal is to check for unsafe content at any intermediate timestep t during generation.

To address this challenge, we present *GenShield*, a framework for unsafe content detection during early generation process. As shown in Figure 2, our GenShield framework considers an undesired concept c and generates paired safe/unsafe latents b_s^t and b_u^t for training purposes (Section 3.2). To achieve safety awareness when generating the above diffusion latents (Section 3.3), we uniquely perform Concept-Content Cross-Attention (C3A) with the concept c and the latents b_s^t and b_u^t, and the resulting output embeddings z_c, z_s, and z_u would be used for our Safety-Aware Contrastive Learning to distinguish unsafe content from the safe ones. For inference, we introduce an early-stage safety assessment mechanism (Section 3.4) that analyzes the semantic features at intermediate diffusion step t, allowing for timely intervention before unsafe content materializes.



(a) Safety-Aware Contrastive Learning (b) Concept-Content Cross-Attention

Figure 2. Overview of our GenShield framework. (a) Given paired safe/unsafe latents b_u^t/b_s^t and the concept c, our GenShield performs Content-Concept Cross-Attention with Safety-Aware Contrastive Learning for injecting safety awareness into diffusion latents. (b) Detailed architecture of the Concept-Content Cross-Attention (C3A) module, where the concept c serve as queries while diffusion latents b_u^t/b_s^t conditioned with timestep embeddings h_t serve as keys and values.

3.2. Paired data generation for safe-unsafe concepts

We now detail how we create diffusion latent training pairs from each safety concept for realizing our GenShield framework. In *GenShield*, we develop a paired data generation pipeline that aims to create semantically aligned safe-unsafe latent pairs conditioned on the concept. As illustrated in Figure 3, we first generate paired prompts using LLMs and then extract their corresponding latent features through the generation process of diffusion models. Specifically, given an unsafe concept (*e.g.*, sexual) and an unsafe instruction I_u, the LLM first generates an unsafe prompt p_u (*e.g.*, "A naked person"). Similarly, the LLM generates its safe counterpart p_s (*e.g.*, "A dressed person") by taking the unsafe prompt p_u and a safe instruction I_s as input:

$$\begin{aligned} p_u &= \text{LLM}(c, I_u), \\ p_s &= \text{LLM}(p_u, I_s). \end{aligned} \tag{1}$$

While the above text prompts could be directly used to train a text-based classifier, rephrased prompts like "An unadorned woman" may still bypass detection. Instead, we extract their visual features using the diffusion model to effectively captures the visual manifestations of unsafe content. That is, we feed these prompt pairs p_u and p_s into the diffusion model to obtain paired diffusion latents b_u^t and b_s^t. To ensure this training pair shares similar visual context, both safe and unsafe latents are generated from the same random noise, and we collect all the intermediate latents from the diffusion U-Net for each timestep t. Such collected latents capture the semantic evolution of both safe and unsafe content throughout the generation process and therefore are suitable to serve the inputs for the following contrastive learning, as detailed in Section 3.3.

3.3. Injecting safety awareness into diffusion latents

With the safe-unsafe training pairs obtained, we now introduce how we perform Safety-Aware Contrastive learning to achieve unsafe content detection. As shown in Figure 2(a), given the concept c and the associated positive and negative sample b_u^t and b_s^t, our GenShield first introduces a *Concept-Content Cross-Attention (C3A)* mechanism that maps both concept embeddings and visual latents into a unified semantic space for contrastive learning. :

$$z_c, z_{u,s} = \text{C3A}(\text{CLIP}(c), \mathbf{b}_{u,s}^t), \tag{2}$$

where z_c and $z_{u,s}$ are the embeddings for the concept of interest and the unsafe/safe content, respectively.

Since the diffusion latents are derived from different timestep t, we employ a Feature-wise Linear Modulation (FiLM) [30] layer inside our C3A module to encode time information into the diffusion latent b_u^t and b_s^t before attention, as shown in Figure 2(b). To be more precise, FiLM applies a timestep-conditioned transformation on the diffusion latents: $\hat{\mathbf{b}}_{u,s}^t = Linear(h_t) \odot \mathbf{b}_{u,s}^t + Linear(h_t)$, where h_t is the timestep embedding from the diffusion model denoising step t. Then, the attention would be performed by taking the textual concept c as query and the transformed visual latents $\hat{\mathbf{b}}_{u,s}^t$ as keys and values.

$$\text{Attention}(\text{CLIP}(c), \text{K} = \hat{\mathbf{b}}_{u,s}^t, \text{V} = \hat{\mathbf{b}}_{u,s}^t). \tag{3}$$

The above framework is optimized through supervised contrastive learning [17] to achieve concept-to-content feature alignment. Formally, we construct triplets consisting of an anchor concept embedding z_c, a positive example from unsafe content z_u, and a negative example from safe content z_s. The resulting positive pair (z_c, z_u) represents

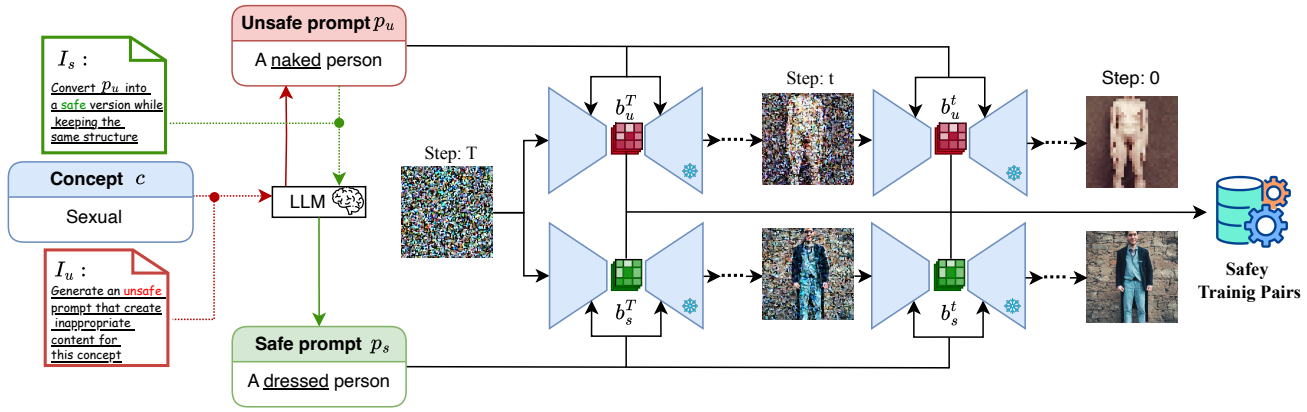


Figure 3. Paired data generation pipeline for collecting safe/unsafe diffusion latents. Given a safety concept c and corresponding unsafe instruction I_u, LLM first generates an unsafe prompt p_u. Similarly, given the safe instruction I_s, LLM converts p_u into a safe version p_s while preserving the semantic structure. Both p_u and p_s are fed into the diffusion model with the same initial noise to extract paired latent features b_u^t and b_s^t at each timestep t, forming safe/unsafe training pairs.

aligned concept-unsafe content, while negative pairs of unsafe and safe representations (z_c, z_s) and (z_u, z_s) should maintain separated. As a result, the contrastive loss $\mathcal{L}$ is formulated as:

$$\mathcal{L} = \log \frac{e(z_u \cdot z_c/\tau)}{e(z_u \cdot z_c/\tau) + e(z_s \cdot z_c/\tau) + e(z_u \cdot z_s/\tau)}, \quad (4)$$

where τ denotes the temperature parameter controlling the distribution concentration and $e(\cdot)$ denotes the exponential function

It is clear that the above loss function would pull unsafe content closer to corresponding concept embeddings, while pushing safe content away. When handling multi-concept settings, we extend the contrastive loss by adding repulsion terms between different unsafe categories, pushing their corresponding embeddings apart to ensure clear separation between different types of unsafe content. With the above scheme, we are able to inject safety awareness into the diffusion latents, allowing us to perform safety detection during the generation process.

3.4. Inference: early-stage safety assessment

During inference, we analyze the latents at each timestep t through the C3A mechanisms for concept detection. For each unsafe concept c in $\mathcal{C}$, C3A takes the current latent features $\mathbf{b}_t$ and concept embedding as input:

$$z, z_c = \text{C3A}(\text{CLIP}(c), \mathbf{b}_t). \quad (5)$$

In *GenShield*, we compute the cosine similarity between z and the concept embedding z_c to measure the manifestation strength of unsafe content. The generation process is halted if the similarity score of any concept exceeds a predefined threshold γ, which can be either determined by the user, or through empirical validation on pre-collected data.

It is worth noting that, while our framework primarily focuses on proactive safety detection during the generation process, it can be adapted for existing image analysis through DDIM inversion [41]. By applying DDIM inversion to transform existing images back to the noise space, we can access latent features $\mathbf{b}_t$ at each timestep during the inversion process. Thus, detection can also be conducted on existing images besides diffusion generation, effectively extending GenShield's applicability to both generation-time and post-generation safety assessment scenarios.

4. Experiments

4.1. Datasets and implementation details

Training datasets. We construct our training dataset using the paired data generation strategy described in Section 3.2, where LLM is used to generate paired safe-unsafe prompts which are then fed through the diffusion model to obtain paired latent features. Specifically, we train our model on seven safety concepts: `self-harm`, `violence`, `shocking content`, `hate`, `harassment`, `sexual` and `illegal activities`. Additionally, we incorporate captions from the MSCOCO dataset as weak safe samples to enhance the model's understanding of common safe content. These captions provide diverse examples of everyday, benign descriptions, helping the model better distinguish general safe content from potentially harmful ones.

Implementation details. All experiments were conducted on an NVIDIA A4500 GPU with 20GB memory, implemented using the PyTorch framework. We employ Stable Diffusion v1.5 as our primary target model for evaluation. The model is configured with a guidance scale of 7.5, using 50 inference steps with DDIM sampler. All images are gen-

	Method	Adversarial Prompts						
		Sneaky Prompt [45]	MMA-Diffusion [43]	I2P-Sexual [39]	I2P [39]	Ring-A-Bell [42]	P4D [7]	AVG.
AUROC ↑	Q16 [38]	74.18	69.00	72.04	72.69	75.31	77.18	73.40
	SD safety checker [8]	79.75	82.57	68.39	57.55	97.89	88.41	79.09
	NudeNet [2]	83.85	90.88	78.48	61.76	92.53	89.79	82.88
	OpenAI-Moderation [28]	*98.50*	73.02	91.93	84.60	99.35	95.68	90.51
	LatentGuard [23]	69.10	88.64	63.61	65.46	82.88	83.52	75.53
	GuardT2I [44]	97.86	*98.86*	*93.05*	**92.56**	*99.91*	**98.36**	*96.77*
	GenShield (Ours)	**99.31**	**99.14**	**95.05**	*92.06*	**99.99**	**99.24**	**97.47**
AUPRC ↑	Q16 [38]	70.81	61.10	71.55	74.04	73.60	75.79	71.19
	SD Safety Checker [8]	89.87	90.98	83.63	65.21	98.95	94.21	87.14
	NudeNet [2]	81.47	87.26	78.63	39.22	91.59	88.62	77.80
	OpenAI-Moderation [28]	*98.48*	58.99	92.14	83.39	8.21	94.87	87.68
	LatentGuard [23]	67.86	90.39	60.83	62.75	75.95	79.91	72.95
	GuardT2I [44]	98.28	*98.95*	89.64	*91.66*	*99.92*	*98.51*	*96.16*
	GenShield (Ours)	**99.36**	**99.84**	**94.23**	**91.76**	**99.99**	**99.49**	**97.45**
FPR@TPR95 ↓	Q16 [38]	71.36	67.60	80.35	82.68	59.57	67.33	71.48
	SD Safety Checker [8]	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	NudeNet [2]	100.00	27.07	100.00	100.00	33.68	33.77	65.75
	OpenAI-Moderation [28]	*4.40*	40.20	35.50	59.09	0.70	*25.42*	27.55
	LatentGuard [23]	90.00	59.00	89.17	88.86	44.21	55.63	71.15
	GuardT2I [44]	6.50	*6.59*	*25.50*	**34.96**	*0.35*	41.67	*19.26*
	GenShield (Ours)	**3.00**	**1.70**	**24.42**	*35.04*	**0.00**	**2.00**	**11.03**
ASR ↓	ESD [10]	28.57	66.7	36.25	-	98.60	79.16	-
	SD Safety Checker [8]	40.50	34.27	62.69	84.36	4.21	23.18	*41.53*
	LatentGuard [23]	61.50	21.10	70.03	*69.85*	38.95	41.72	50.52
	GuardT2I [44]	*9.89*	*10.20*	*26.4*	-	*3.16*	*8.75*	-
	GenShield (Ours)	**5.00**	**3.00**	**24.69**	**40.00**	**0.00**	**6.67**	**13.22**

Table 1. Quantitative comparisons when defending against different types of prompt-based attack. All values are reported in percentage (%). **Bolded** values are the highest performance, with the *underlined italicized* as the second highest ones.

erated at 512 × 512 resolution, with 5 syntheses generated per prompt. For training, we use the Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 512, training for 1 hour.

4.2. Generation-time defense

4.2.1. Defense against prompt-based attack

Evaluation setup. For evaluation metrics, following established practices in [25, 32, 44], we employ AUROC to measure the model's ability to distinguish between adversarial and normal prompts across different thresholds, AUPRC to evaluate the precision-recall trade-off, and FPR@TPR95 to measure the False Positive Rate when True Positive Rate is fixed at 95%. Higher values of AUROC and AUPRC indicate better performance, while lower FPR@TPR95 is preferred. We also report the Attack Success Rate (ASR), which represents the percentage of adversarial prompts that successfully bypass the safety system. Following GuardT2I [44], we conduct experiments to evaluate our method against various types of adversarial prompts.

The test datasets are collected from six sources: SneakyPrompt [45], MMA-Diffusion [43], I2P Dataset [39], Ring-A-Bell [42], and P4D [7]. While most of these datasets primarily focus on NSFW sexual content, the I2P Dataset provides human-written prompts across seven diverse safety categories: `self-harm`, `violence`, `shocking content`, `hate`, `harassment`, `sexual` and `illegal activities`. Additionally, as in GuardT2I, we augment the I2P dataset with 200 additional sexual-themed prompts, creating an expanded subset

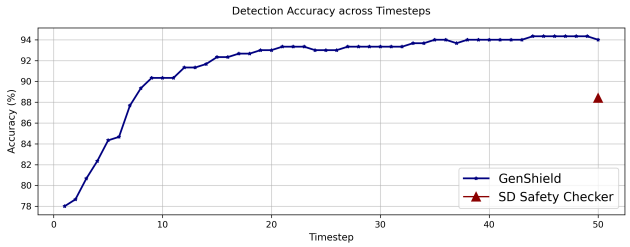


Figure 4. Detection accuracy against prompt-based attack (P4D [7]) on various diffusion timesteps. It can be seen that we can achieved comparable accuracy as that of post-hoc methods like SD safety checker [8] in the early stage of generation.

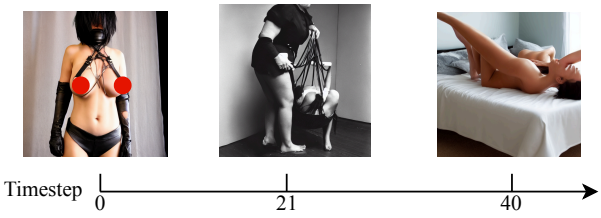


Figure 5. Example unsafe images detected at each timesteps. It suggests images explicitly related to the unsafe concept can be easily detected in early diffusion stages.

referred to as I2P-Sexual.

Compared methods. We evaluate against three categories of safety approaches. For model fine-tuning methods, we include ESD [10]. For text-based classifiers, we include OpenAI Moderation [28], GuardT2I [44], and Latent-Guard [23]. For post-hoc safety checkers that analyze generated images, we compare against Q16 [38], the built-in Stable Diffusion safety checker [8], and NudeNet [2].

Discussions and analysis. The experimental results in Table 1 validate the effectiveness of GenShield against various adversarial prompts. Our framework achieves substantial detection performance with an average AUROC of 97.47%, while maintaining high precision-recall characteristics as evidenced by the 97.45% AUPRC score. Most significantly, GenShield demonstrates strong capability in minimizing false positives, achieving an average FPR@TPR95 of only 11.03%. The framework's performance remains stable across different attack scenarios, with particularly strong results on sophisticated attacks like Ring-A-Bell (99.99% AUROC) and MMA-Diffusion (99.14% AUROC), indicating robust detection capabilities. Further supporting these results, GenShield achieves an average Attack Success Rate (ASR) of 13.22%, substantially outperforming other methods and demonstrating its effectiveness in preventing successful adversarial attacks.

Method	AUROC ↑	AUPRC ↑	FPR@TPR95 ↓
Safety Checker [8]	59.02	79.51	100.00
Q16 [38]	58.00	56.42	90.00
NSFW-detector [19]	72.95	64.89	54.10
GenShield (Ours)	**93.61**	**89.79**	**15.00**

Table 2. Quantitative comparisons when defending against image-based attack in MMA-Diffusion [43].

Accuracy versus timesteps. We analyze GenShield's detection accuracy across timesteps with a detection threshold of 0.87. As shown in Figure 4, the accuracy rises sharply from 78% to 90% in early stages (timesteps 0-10), then gradually stabilizes at 93% after timestep 20. This demonstrates that GenShield is highly responsive, achieving robust detection at just 20% of the total diffusion steps, enabling early termination of unsafe content generation without completing the full 50-step process. Additionally, compared to post-hoc methods, our GenShield achieves superior performance. We further provide the visualization of detected images at different timesteps, as shown in Figure 5. In earlier timesteps (0-20), the framework demonstrates robust performance in identifying clearly problematic content with high detection accuracy. As the diffusion process progresses to intermediate timesteps (21-40), GenShield encounters more nuanced cases requiring detailed analysis. In later timesteps (40-50), the framework faces the most challenging scenarios where content ambiguity is highest, yet successfully maintains its ability to identify potentially unsafe content even when the visual presentation is contextually ambiguous.

4.2.2. Defense against image-based attack

Beyond text-based attacks, we also evaluate GenShield against MMA-Diffusion's image modality attack [43], which targets post-hoc safety checkers by adding adversarial perturbations to the inpainting process, generating restricted content while evading detection. As shown in Table 2, GenShield achieves substantially better detection performance across all metrics compared to existing safety checkers. This robustness stems from our approach of analyzing the generation process itself, effectively preventing the adversarial manipulations from materializing in the final output.

4.3. Post-generation safety assessment

Evaluation setup. We evaluate GenShield's capability in detecting diverse categories of unsafe content on the UnsafeBench [33] dataset, which contains both real-world and AI-generated unsafe images. Since UnsafeBench provides images rather than prompts, we utilize DDIM inversion process to obtain their latent representations for

Method	Hate	Harassment	Violence	Self-Harm	Sexual	Shocking	Illegal Activity	AVG.
Q16 [38]	*0.533*	0.475	0.648	*0.483*	-	0.784	0.571	-
MultiHeaded [32]	0.292	-	0.426	-	0.757	0.749	-	-
SD Safety Checker [8]	-	-	-	-	0.785	-	-	-
NSFW_Detector [19]	-	0.449	-	-	0.738	-	-	-
NudeNet [2]	-	-	-	-	0.624	-	-	-
LLaVA* [22]	0.210	0.189	0.550	**0.590**	0.780	0.800	0.519	0.520
InstructBLIP* [9]	0.270	0.368	0.615	0.333	0.777	0.697	0.687	0.535
GPT-4V* [1]	0.423	**0.681**	*0.738*	**0.590**	*0.847*	*0.839*	*0.780*	*0.700*
GenShield (Ours)	**0.593**	*0.644*	**0.766**	0.459	**0.883**	**0.850**	**0.815**	**0.716**

Table 3. Quantitative comparisons of post-generation safety assessment on the UnsafeBench dataset [33]. All values are reported in F1-Score. **Bolded** values are the highest performance. The *underlined italicized* values are the second highest performance. * marks VLM-based classifiers.

Model	Time (ms) ↓	#Params (M) ↓	AUROC ↑
Safety Checker [8]	2630.53	290	79.09
LatentGuard [23]	11.24	1	75.53
GuardT2I [44]	-	538	96.77
GenShield (Ours)	87.10	11	97.47

Table 4. Efficiency comparisons of inference time and the number of parameters among different safety detection methods.

our detection framework. To better handle the domain differences between AI-generated and real-world images, we incorporate UnsafeBench's training set into our training process. The inclusion of real-world images alongside AI-generated content makes this benchmark particularly valuable for evaluating detection systems under realistic conditions. Following standard practices, we use F1-score as the primary metric for each category.

Discussions and analysis. The quantitative results in Table 2 demonstrate the effectiveness of GenShield across multiple safety categories on the UnsafeBench dataset. While our framework shows relatively moderate performance in self-harm detection (0.459) compared to GPT-4V (0.590), what makes these results particularly significant is that GenShield achieves a higher average F1-score of 0.716 compared to GPT-4V's 0.700, despite using a substantially lighter architecture with significantly fewer parameters. The framework shows particular strength in challenging categories such as hate speech detection with an F1-score of 0.593, considerably higher than GPT-4V's 0.423. This performance validates our key insight that analyzing diffusion latents with targeted safety-aware training provides an effective approach for unsafe content detection that balances efficiency with detection capability.

4.4. Efficiency comparisons

Table 4 presents the efficiency comparison of different safety detection methods. Our GenShield framework achieves competitive efficiency with only 10.56M parameters, which is significantly smaller than SafetyChecker (1306M) and GuardT2I (538M). While pure text-based approaches achieve faster inference times, the performance gap in latency remains within the same order of magnitude, and detection accuracy is typically prioritized over such minor speed differences in real-world safety applications.

5. Conclusion

In this work, we present GenShield, a novel safety detection framework for text-to-image diffusion models that leverages diffusion latent features to identify unsafe content during the generation process. To facilitate the learning of our GenShield framework, we design a data generation strategy to produce safe-unsafe training pairs from LLMs and diffusion models. Given such training pairs together with the associated concepts as inputs, our Safety-Aware Contrastive Learning employs Concept-Content Cross-Attention (C3A) mechanism to achieve the alignment between concept embeddings and diffusion latents. With the above designed learning scheme, our GenShield framework would inject safety awareness into the diffusion latents, enabling unsafe content detection during the ongoing generation process. Through experiments, we demonstrate that GenShield achieves superior performance in both generation-time defense and post-generation safety assessment, while maintaining high efficiency with our lightweighted design. More importantly, GenShield demonstrates effectiveness against various adversarial prompts and efficiency through early-stage intervention, achieving both robustness and responsiveness for text-to-image safety moderation.